

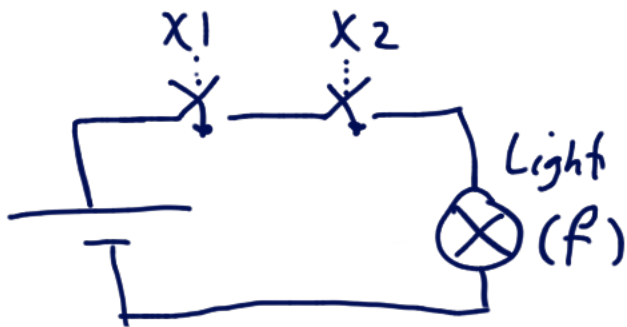
Logic & gates

①

Transistor → switch $\begin{cases} \text{ON} = 1 \\ \text{OFF} = 0 \end{cases}$



① Logic AND (series)



AND

$$\left. \begin{array}{l} f(x_1, x_2) = x_1 \cdot x_2 \\ = x_1 x_2 \end{array} \right\} \begin{array}{l} \text{function} \\ \text{output} \end{array} \quad \begin{array}{l} \text{variables} \\ \text{input} \end{array}$$

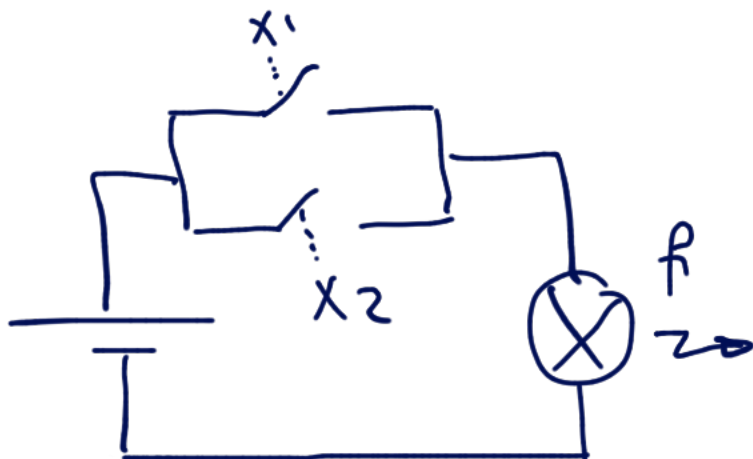
AND gate



truth table

x_1	x_2	f
0	0	0
0	1	0
1	0	0
1	1	1

② Logic OR (parallel)



$$f(x_1, x_2) = x_1 + x_2$$

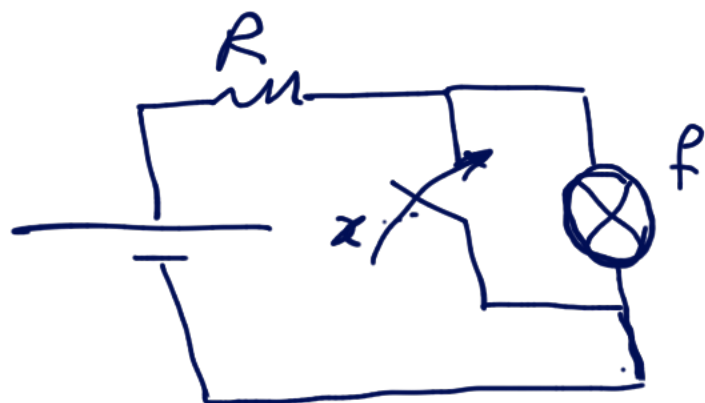
Or



x_1	x_2	f
0	0	0
0	1	1
1	0	1
1	1	1

③ Logic NOT (inversion = complement)

②



$$x=0 \longrightarrow f=1 \text{ (on)}$$

$$x=1 \longrightarrow f=0 \text{ (off)}$$

$$\left. \begin{aligned} f(x) &= \bar{x} \\ &= \sim x \\ &= x' \end{aligned} \right\} \text{NOT}$$



x	f
0	1
1	0

④ NAND gate (AND-NOT)

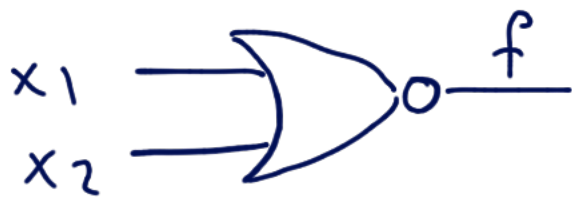


$$\begin{aligned} f(x_1, x_2) &= \overline{x_1 \cdot x_2} \\ &= (x_1 \cdot x_2)' \end{aligned}$$

x ₁	x ₂	f
0	0	1
0	1	1
1	0	1
1	1	0

⑤ NOR gate

③

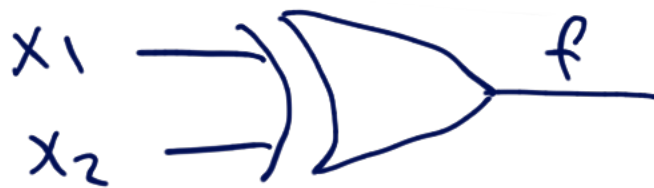


$$f(x_1, x_2) = \overline{x_1 + x_2}$$

$$= (x_1 + x_2)'$$

x_1	x_2	f
0	0	1
0	1	0
1	0	0
1	1	0

⑥ XOR gate (exclusive OR)



$$f(x_1, x_2) = x_1 \oplus x_2$$

$$= \overline{x_1}x_2 + x_1\overline{x_2}$$

different

x_1	x_2	f
0	0	0
0	1	1
1	0	1
1	1	0

⑦ XNOR gate (XOR - NOT)



$$f(x_1, x_2) = x_1 \odot x_2$$

$$= \overline{x_1 \oplus x_2}$$

$$= x_1.x_2 + \overline{x_1}.\overline{x_2}$$

Similar

x_1	x_2	f
0	0	1
0	1	0
1	0	0
1	1	1

Truth table :

is used to show the behaviour of a logical expression or circuit (network of gates).

n : number of inputs (variables)

$$\# \text{ of rows} = 2^n$$

$$n = 2 \longrightarrow \text{rows} = 2^2 = 4$$

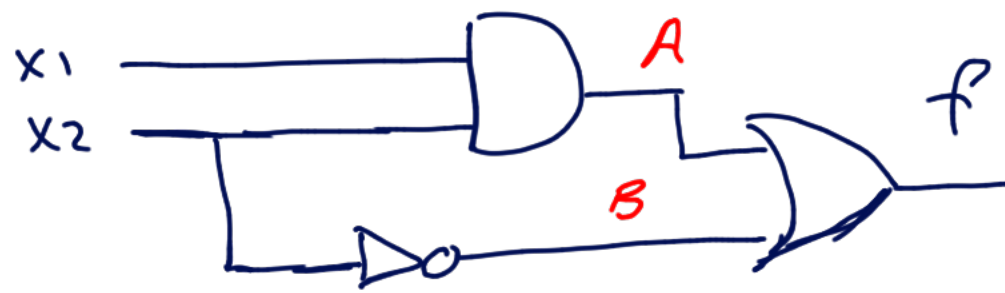
$$n = 3 \longrightarrow \text{rows} = 2^3 = 8$$

4 x_1	2 x_2	1 x_3	f
0	0	0	
0	0	1	
0	1	0	
0	1	1	
1	0	0	
1	0	1	
1	1	0	
1	1	1	

Example I

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Given the circuit



=>

- 1- Derive the logical expression (Function) for the circuit.
- 2- Draw the truth table.
- 3- What is the cost of the circuit?
- 4- Draw the timing diagram.

① function

$$f(x_1, x_2) = A + B$$

$$= x_1 \cdot x_2 + B$$

$$f = x_1 \cdot x_2 + \overline{x_2}$$

② truth table

x_1	x_2	A $x_1 \cdot x_2$	B $\overline{x_2}$	f
0	0	0	1	1
0	1	0	0	0
1	0	0	1	1
1	1	1	0	1

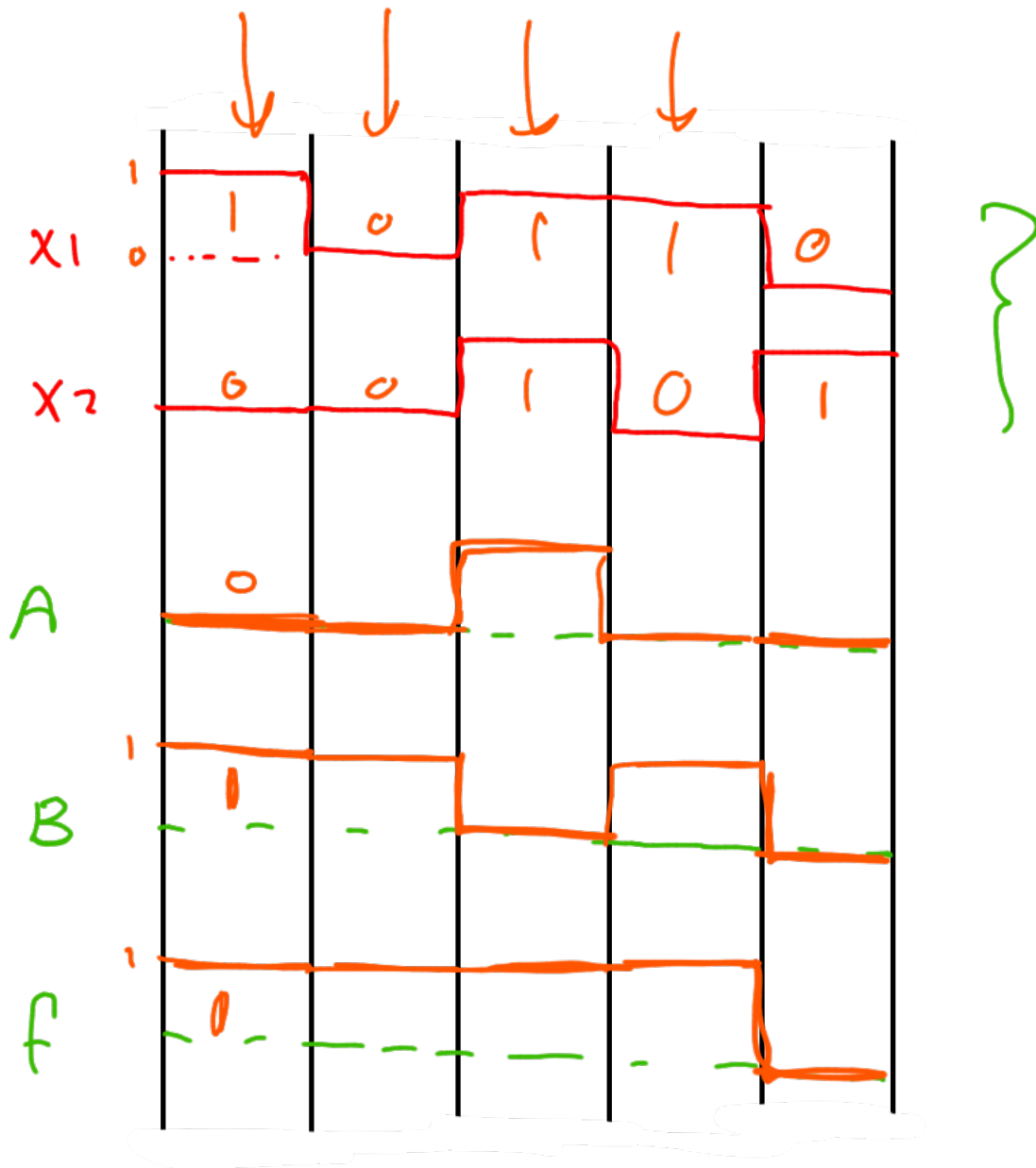
③ $\text{cost} = \# \text{ of gates} + \# \text{ of inputs (each gate)}$

⑥

$$\text{cost of } f = 3 + 5 = 8$$

(Fan-in)

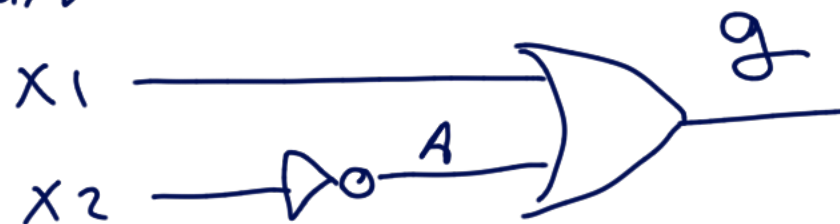
④ Timing diagram (It is best to extract it from the truth table.)



Example II

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Given the circuit



- 1- Derive the logical expression (Function) for the circuit.
- 2- Draw the truth table.
- 3- What is the cost of the circuit?
- 4- Draw the timing diagram

1- Function

$$g(x_1, x_2) = x_1 + A$$

$$g = x_1 + \bar{x}_2$$

2- Truth table

x_1	x_2	A $\bar{x}_2$	g
0	0	1	1
0	1	0	0
1	0	1	1
1	1	0	1

3- cost of $g = 2 + 3 = 5$

Compare the circuits in examples I and II (considering the truth tables).

Note that $f = g$

The two circuits are functionally equivalent
but g is better due to the cost

prove $\rightarrow f = g$

$$x_1 \cdot x_2 + \overline{x_2} = x_1 + \overline{x_2}$$

We can prove that two circuits (or two logical expressions) are functionally equivalent in two ways:

- 1- Truth tables
 - 2- Boolean algebra
-

- from truth tables we note that $f = g$.
- this means that $x_1 \cdot x_2 + x_2' = x_1 + x_2'$
- in the following pages we can also prove it using Boolean algebra

Boolean algebra

① Axioms :

$$0 \cdot 0 = 0$$

$$0 \cdot 1 = 1 \cdot 0 = 0$$

$$1 \cdot 1 = 1$$

$$0 + 0 = 0$$

$$0 + 1 = 1 + 0 = 1$$

$$1 + 1 = 1$$

$$\overline{0} = 1$$

$$\overline{1} = 0$$

② One variable

$$x \cdot 0 = 0 \xrightarrow{\text{dual}} x + 1 = 1$$

$$x \cdot 1 = x$$

$$x + 0 = x$$

$$x + 1 = 1$$

$$x \cdot x = x$$

$$x + x = x$$

$$x \cdot \overline{x} = 0$$

$$x + \overline{x} = 1$$

$$\overline{\overline{x}} = x$$

III 2 or more variables

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① $x \cdot y = y \cdot x$ $\longrightarrow$ Commutative
 $\downarrow$ dual $x + y = y + x$

② $x \cdot (y \cdot z) = (x \cdot y) \cdot z$

③ $x \cdot (y + z) = x \cdot y + x \cdot z$

④ $x + x \cdot y = x$
 $\downarrow$ dual $x \cdot (x + y) = x$

⑤ $x \cdot y + x \cdot \bar{y} = x$
 $\downarrow$ dual $(x + y) \cdot (x + \bar{y}) = x$

⑥ $x + \bar{x} \cdot y = x + y$
 $\downarrow$ dual $x \cdot (\bar{x} + y) = x \cdot y$

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⑦ $\overline{x \cdot y} = \bar{x} + \bar{y}$ De Morgan's theorem

$$(8) \quad \underline{x.y} + y.z + \underline{\bar{x}.z} = x.y + \bar{x}.z$$

* Principle of Duality

Given a logical expression, its dual is obtained

By replacing all + (ors) with . (Ands) and vice versa, and

By replacing all 0s with 1s and vice versa.

$$\bar{y} + y\bar{x} \stackrel{(6)}{=} \bar{y} + \bar{x}$$

$$f = x_1 \cdot x_2 + \bar{x}_2$$

$$g = x_1 + \bar{x}_2$$

$$x_1 \cdot x_2 + \bar{x}_2 = x_1 + \bar{x}_2$$



prove

$$\frac{x\bar{z} + \bar{y}\bar{z} + xz + \bar{y}z}{LHS} = \frac{\bar{x}\bar{y} + xy + x\bar{y}}{RHS}$$

$$\begin{aligned} LHS &= \underbrace{x\bar{z} + \bar{y}\bar{z}}_{x + \bar{y}} + \underbrace{xz + \bar{y}z}_{x + \bar{y}} \\ &= x + \bar{y} \end{aligned}$$

$$\begin{aligned} RHS &= \bar{x}\bar{y} + \underbrace{xy + x\bar{y}}_x \\ &= \bar{x}\bar{y} + x \\ &= x + \bar{y} = LHS \end{aligned}$$

